Text Summarizer

Rahul Shah from IITDelhi

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1 Introduction

Text summarization involves creating a shorter version of a text that retains its key information. While humans are able to understand the main points of a text by simply reading it, machines can also perform this task through the use of natural language processing (NLP). There are various applications of automatic text summarization, including condensing customer reviews, summarizing news articles, and creating summary reports from business meeting notes. In this Notebook, we will explore the details of this exciting application of NLP in greater depth.

Definition:
Natural Language Processing (NLP) is a field of Artificial Intelligence that focuses on the interaction between computers and human (natural) languages.
Key Goals:

- Understand and interpret human language
- Process and extract useful information
- Generate meaningful responses

Common NLP Tasks:

- Text Classification (e.g., spam detection)
- Sentiment Analysis
- Named Entity Recognition (NER)
- Machine Translation
- Speech Recognition
- Question Answering

Techniques Used:

- Rule-based methods
- Statistical models
- Deep learning: RNNs, LSTMs, Transformers (e.g., BERT, GPT)

Applications:

- Voice assistants (Siri, Alexa)
- Search engines (Google)
- Chatbots
- Language translators
- Grammar correction tools

2 Methods

There are mainly 2 methods of summarizing:
Extractivesummarization involves identifying and extracting the most important sentences from the original text. Extractive summarization is easier to implement and can be done quickly using an unsupervised approach that does not require prior training.
Abstractive involves understanding the main ideas of the text and generating a new, summarized version based on that understanding. Abstractive summarization has the advantage of being able to generate new text, but it is more complex to implement and requires language generation capabilities.
Note:We will be focusing on abstracting summarization.

Extractive summarization methods

The Extractive based summarization method selects informative sentences from the document as they exactly appear in source based on specific criteria to form summary. The main challenge before extractive summarization is to decide which sentences from the input document is significant and likely to be included in the summary. For this task, sentence scoring is employed based on features of sentences. It first, assigns a score to each sentence based on feature then rank sentences according to their score. Sentences with the highest score are likely to be included in final summary. Following methods are the technique of extractive text summarization.
1.TF-IDF
Term frequency (TF) and the inverse document frequency (IDF) are numerical statistics presents how important a word in a given document. TF is number of times a term occurs in the document and IDF is a measure that diminishes the weight of terms that occur very frequently in the collection and increases the weight of terms that occur rarely. Then sentences are scored according to product and sentence having high score are included in summary. One problem with this method is sometimes longer sentences gets high score due to fact that they contain more number of words
2.Cluster based methods
Documents are composed in such a manner that they address different ideas in separate sections. It is natural to think that summaries should address different themes separated into sections of the document. In case that the document for

which summary is being delivered is of entirely different subjects then summarizer assimilates this aspect through clustering. The document is represented using TF-IDF of scores of words. High frequency term represents the theme of a cluster. Sum- mary sentence is selected based on relationship of sentence to the theme of cluster. Cluster based method generate summary of high relevance, to the given query or document topic.

3.Text summarization with nueral networks

A Neural Network is a processing system modeled on the human brain that tries to reenact its learning process. Neural network is an interconnected assembly of artificial neurons that uses a numerical model of computation for data processing. In case of text summarization, the strategy includes preparing the neural systems to capture the sort of sentences that ought to be incorporated into the summary. Neural Network is trained with sentences in test paragraph where each sentence is checked as to be included in summary or not. Training is done in accordance with the need of user. Neural network accurately classifies summary sentences but faces the problem of excessive training time.

4.Text Summarization with Fuzzy Logic Fuzzy Logic is a way of reasoning that resembles with the human reasoning which based on degrees of truth rather than the usual true or false (1 or 0) Boolean logic. The fuzzy system is designed with fuzzy rules and membership function which highly affect the performance. A value from zero to one is obtained for each sentence in output based on feature contained in sentence and rules defined in a knowledge base. Important sentences are extracted using IF-THEN rules based on feature criteria. Sentences are ranked in order according to score. In summary, sentence having high score are extracted. Fuzzy logic systems are simple and flexible can take imprecise, distorted, noisy input information.

5.Graph based Method

In this method every sentence of the document is considered as a vertex of the graph. Sentences are connected with an edge if there exist common semantic relation and based on this relation connecting edge is given weight. A graph based ranking algorithm is used to decide the importance of a vertex within a graph. Vertexes with high cardinality are considered as important sentences and included in summary. Graph based method does not require deep linguistic knowledge, nor do-main knowledge for summarization. Directed graph maintains a flow of text while an edge in undirected graph captures relation using co-occurrence of terms.

6.Latent Semantic Analysis Method

LSA is algebraic statistical method that extracts meaning and resemblance of a sentence by the information about words in a particular environment. It keeps information about which words are used in sentence and reserve information of common word amongst sentences, the more common word between sentences the more it relevant. LSA extracts the source text and converts into term sentence matrix and process it through Singular Value Decomposition (SVD) for finding semantically similar words and sentences. SVD models relationships among words and sentences. The key point of LSA is it avoids the problem of synonyms but it uses only information in the input text and does not use the information of word order, syntactic relations is the major limitation of this method.

7.Machine Learning approach

In this method, the training data set is used for reference to generate summary. Summarization process is modeled as a classification problem. Sentences are classified as summary sentences and non-summary sentences based on the features that they possess.

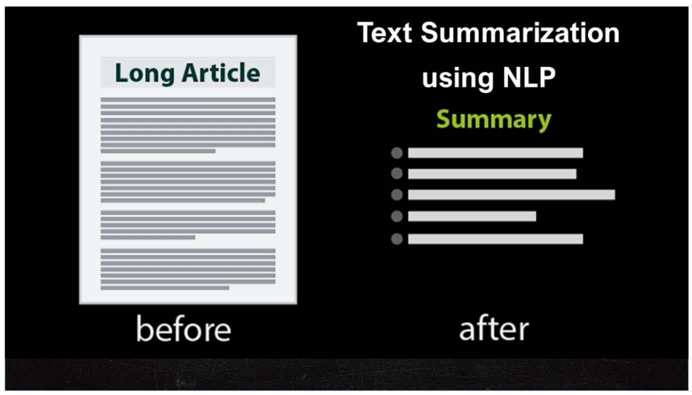


Figure 1: Example

Remark:we will be focusing on graph based methods.

Graph Based method understanding

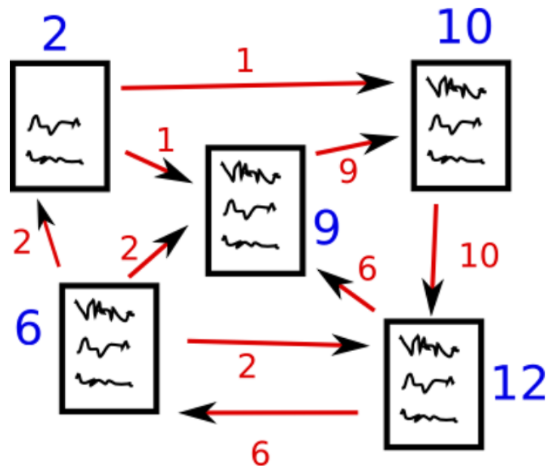


Figure 2: Graph method

The core idea behind this method is to find the similarities among all the sentences and returning the sentences having maximum similarity scores. We use Cosine Similarity as the similarity matrix and TextRank algorithm to rank the sentences based on their importance.
Before understanding the TextRank algorithm, it is important to briefly talk about the PageRank algorithm, the influence behind TextRank.
PageRank is a graph based algorithm used by Google to rank the web pages based on a search result. PageRank first creates a graph with pages being the vertices and the links between pages being the edges. The PageRank score is calculated for each page, which is basically the probability of user visiting that page. Here is a nice paper explaining the PageRank algorithm.
Ranking:
Similarity of TextRank with PageRank can be underlined using following points:
1. sentences are used in place of pages as vertices in graph.
2. Similarity between sentences is used as edges instead of links.
3. Instead of page visit probability, sentences similarities are used to calculate the ranks.
Ranking algorithm generates a graph from the natural language texts. The basic idea of any graph based algorithm is based on 'voting' or 'recommendation'. When one vertex has links to another, it is basically casting a vote for that vertex. Higher the number of votes for a vertex, higher the importance of that vertex. Score of a vertex depend on two factors:
1. Number of votes cast for it.
2. importance of vertices casting votes for it.
Steps to Follow in Ranking:

1. Extract all sentences from the original text.
2.Create vectors for all the sentences based on the tokens (words) present in them.
3. Calculate Cosine similarity between each sentence pair. Create a n x n similarity matrix where n is the number of sentences.
4. Create a graph using the similarity matrix, where each vertex represents a sentence and the edge between two vertices represents similarity.
5. Rank the sentences based on similarity score and return the top N number of sentences to be included in the summarized version.
Remark: A brief note on Cosine similarity will help here . Cosine distance between any two vectors in a multi-dimensional space is calculated using Cosine of the angle between them. The formulae for Cosine distance between vectors Va and Vb can be written as:

$$CosineDistance(Vec_a, Vec_b) = 1 - Cosine(AnglebetweenVec_a, Vec_b)$$

We can say that for similar vectors the Cosine distance will be low and the Cosine similarity will be high.

3 Loading and storing data

About dataset:
This dataset for extractive text summarization has four hundred and seventeen political news articles of BBC from 2004 to 2005 in the News Articles folder. For each articles, five summaries are provided in the Summaries folder. The first clause of the text of articles is the respective title.

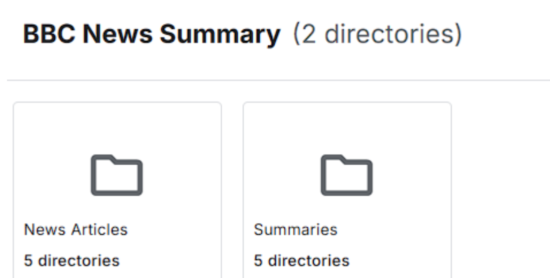
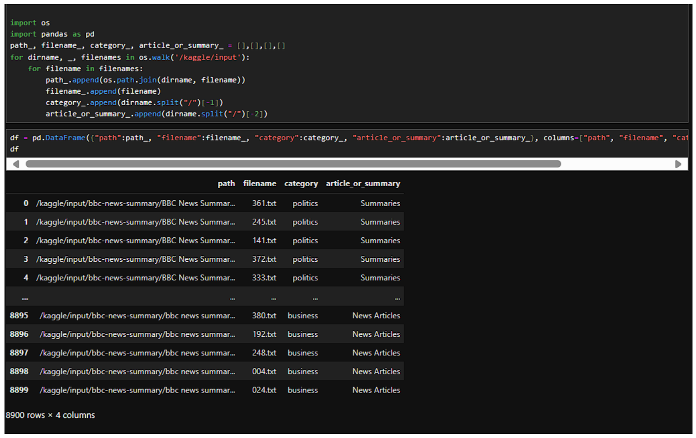


Figure 3: Data-set

BBC Dataset link from kaggle.



```
import os
import pandas as pd
path_, filename_, category_, article_or_summary_ = [],[],[],[]
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        path_.append(os.path.join(dirname, filename))
        filename_.append(filename)
        category_.append(dirname.split("/")[-1])
        article_or_summary_.append(dirname.split("/")[-2])
```

```
df = pd.DataFrame({"path":path_, "filename":filename_, "category":category_, "article_or_summary":article_or_summary_}, columns=["path", "filename", "cat
df
```

	path	filename	category	article_or_summary
0	/kaggle/input/bbc-news-summary/BBC News Summar...	361.txt	politics	Summaries
1	/kaggle/input/bbc-news-summary/BBC News Summar...	245.txt	politics	Summaries
2	/kaggle/input/bbc-news-summary/BBC News Summar...	141.txt	politics	Summaries
3	/kaggle/input/bbc-news-summary/BBC News Summar...	372.txt	politics	Summaries
4	/kaggle/input/bbc-news-summary/BBC News Summar...	333.txt	politics	Summaries
...	...	...	...	...
8895	/kaggle/input/bbc-news-summary/bbc news summar...	380.txt	business	News Articles
8896	/kaggle/input/bbc-news-summary/bbc news summar...	192.txt	business	News Articles
8897	/kaggle/input/bbc-news-summary/bbc news summar...	248.txt	business	News Articles
8898	/kaggle/input/bbc-news-summary/bbc news summar...	004.txt	business	News Articles
8899	/kaggle/input/bbc-news-summary/bbc news summar...	024.txt	business	News Articles

8900 rows × 4 columns

Figure 4: loading dataset

4 Exploratory Data Analysis(EDA)

1.Distribution of Number of Articles in Each Category

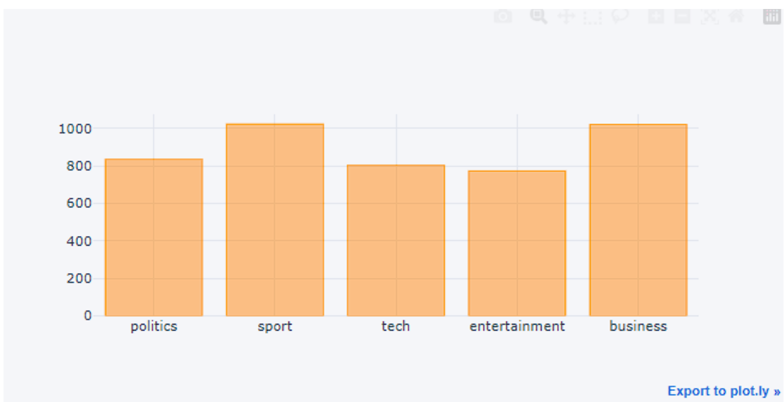


Figure 5: Article v/s category Bar graph

2.Distribution of Category and its Values

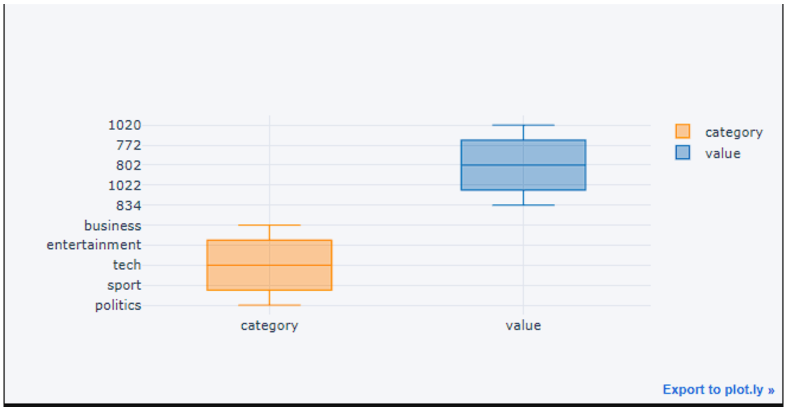


Figure 6: Box plot B/W category and values.

Remark: No outliers found.

3.Distribution Size of Each Category

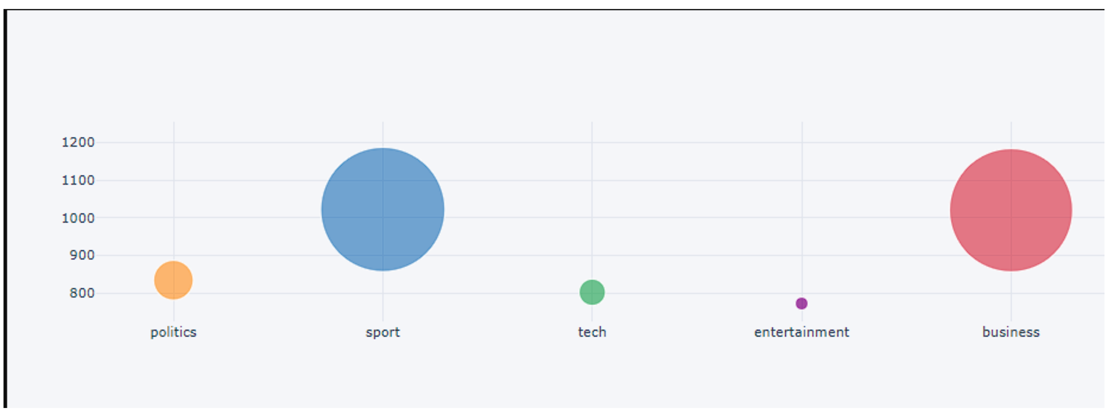


Figure 7: Bubble chart of each category.

4.Coverage Ratio of Each Category

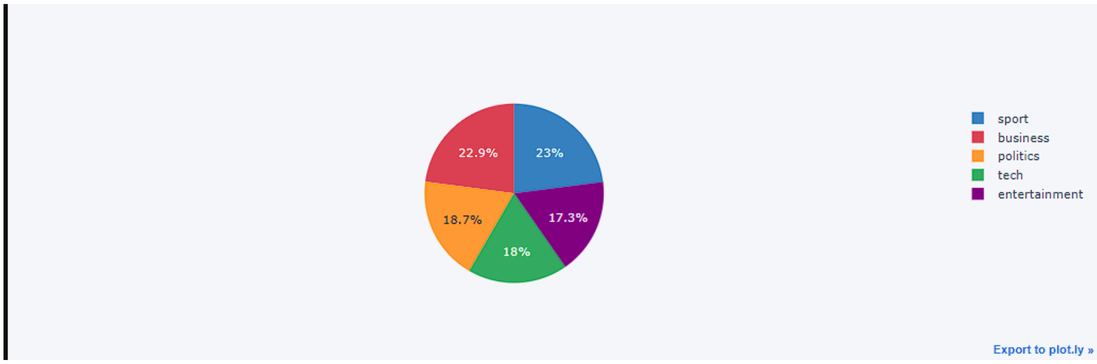


Figure 8: Pie-Chart

5 Preprocessing

Sentence Tokenization:
Tokenization is a fundamental step in Natural Language Processing (NLP). It involves dividing a Textual input into smaller units known as tokens. These tokens can be in the form of words, characters, sub-words, or sentences. It helps in improving interpretability of text by different models.

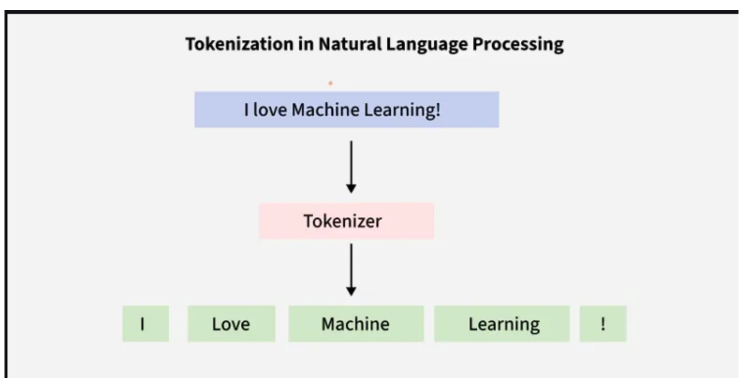


Figure 9: Tokenization

Importance of Tokenization in NLP

Tokenization is an essential step in text processing and Natural Language Processing (NLP) for several reasons:

- **Effective Text Processing:** Reduces the size of raw text, resulting in easy and efficient statistical and computational analysis.

- **Feature Extraction:** Text data can be represented numerically for algorithmic comprehension by using tokens as features in machine learning models.
- **Information Retrieval:** Tokenization is essential for indexing and searching in systems that store and retrieve information efficiently based on words or phrases.
- **Text Analysis:** Used in sentiment analysis and named entity recognition to determine the function and context of individual words in a sentence.
- **Vocabulary Management:** Generates a list of distinct tokens and helps manage a corpus's vocabulary.
- **Task-Specific Adaptation:** Adapts to the needs of particular NLP tasks such as summarization and machine translation.

Figure 10: Sentence Tokenization

```
[8]: import nltk
     from nltk.corpus import stopwords
     from nltk.tokenize import sent_tokenize
     import numpy as np
     import networkx as nx
     import re

[9]: def read_article(text):
         sentences =[]
         sentences = sent_tokenize(text)
         for sentence in sentences:
             sentence.replace("[^a-zA-Z0-9]"," ")
         return sentences

[10]: file_path = df[df['article_or_summary']=='News Articles'].iloc[0]['path']
      with open(file_path, "r") as f:
          article = f.read()

[11]: sent_tok = read_article(article)
      sent_tok

[11]: ["Budget to set scene for election\n\nGordon Brown will seek to put the economy at the centre of Labour's bid for a third term in power when he delivers
     his ninth Budget at 1230 GMT.",
      'He is expected to stress the importance of continued economic stability, with low unemployment and interest rates.',
      'The chancellor is expected to freeze petrol duty and raise the stamp duty threshold from £60,000.',
      'But the Conservatives and Lib Dems insist voters face higher taxes and more means-testing under Labour.',
      'Treasury officials have said there will not be a pre-election giveaway, but Mr Brown is thought to have about £2bn to spare.',
      "- Increase in the stamp duty threshold from £60,000 \n - A freeze on petrol duty \n - An extension of tax credit scheme for poorer families \n - Possi
     ble help for pensioners The stamp duty threshold rise is intended to help first time buyers - a likely theme of all three of the main parties' general e
     lection manifestos.",
      'Ten years ago, buyers had a much greater chance of avoiding stamp duty, with close to half a million properties, in England and Wales alone, selling f
     or less than £60,000.',
      'Since then, average UK property prices have more than doubled while the starting threshold for stamp duty has not increased.',
      "Tax credits As a result, the number of properties incurring stamp duty has rocketed as has the government's tax take.",
      'The Liberal Democrats unveiled their own proposals to raise the stamp duty threshold to £150,000 in February.',
      'The Tories are also thought likely to propose increased thresholds, with shadow chancellor Oliver Letwin branding stamp duty a "classic Labour stealth
     tax".',
      'The Tories say whatever the chancellor gives away will be clawed back in higher taxes if Labour is returned to power.',
      'Shadow Treasury chief secretary George Osborne said: "Everyone who looks at the British economy at the moment says there has been a sharp deterioratio
     n in the public finances, that there is a black hole," he said.',
      '"If Labour is elected there will be a very substantial tax increase in the Budget after the election, of the order of around £10bn."',
      "But Mr Brown's former advisor Ed Balls, now a parliamentary hopeful, said an examination of Tory plans for the economy showed there would be a £35bn d
     ifference in investment by the end of the next parliament between the two main parties.",
      'He added: "I don\'t accept there is any need for any changes to the plans we have set out to meet our spending commitments."',
      'For the Lib Dems David Laws said: "The chancellor will no doubt tell us today how wonderfully the economy is doing," he said.',
      '"But a lot of that is built on an increase in personal and consumer debt over the last few years - that makes the economy quite vulnerable potentially
     if interest rates ever do have to go up in a significant way."',
      'SNP leader Alex Salmond said his party would introduce a £2,000 grant for first time buyers, reduce corporation tax and introduce a citizens pension f
```

Figure 11: Code execution of tokenization

Spell correction

Spell correction is the process of detecting and correcting spelling errors in text.
How Spell Correction Works

- **Dictionary Lookup:** Check words against a predefined vocabulary to identify unknown or invalid words.
- **Edit Distance (Levenshtein Distance):** Measure how many basic operations (insertions, deletions, substitutions) are needed to convert one word into another. Lower distance suggests a more likely correction.
- **Probabilistic Models:** Use contextual probability to select the most likely intended word. A common approach is the *Noisy Channel Model*, which models spelling errors as noise in transmission.
- **Neural Approaches:** Employ deep learning techniques, such as sequence-to-sequence (seq2seq) models or transformers, to generate context-aware corrections based on entire sentence structure.

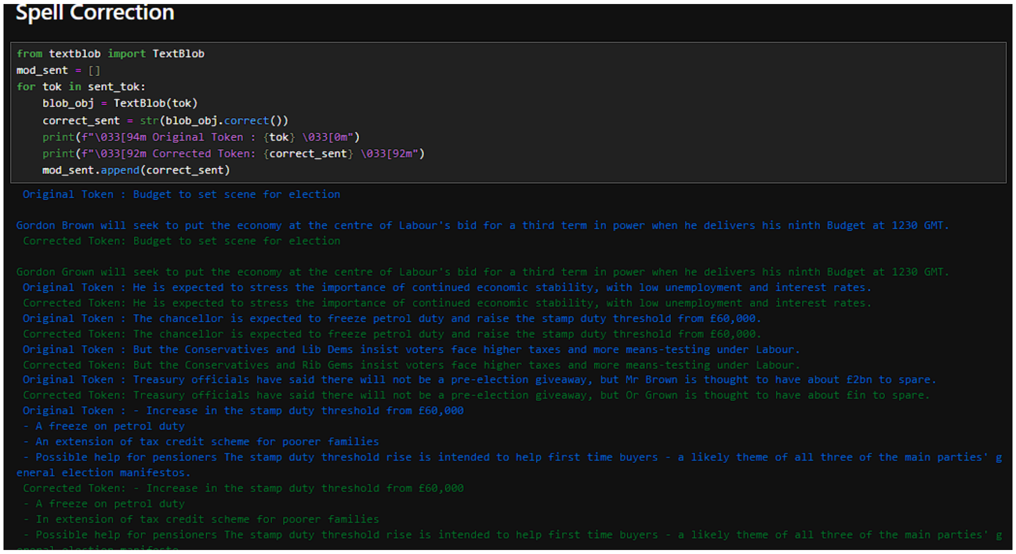


Figure 12: Spell-Correction

Sentence Similarity

Using NLP to Tokenise sentences and storing the values to a iteratable lists and each of these sentence will be encoded using Google Universal sentence Encoder(GSE) or we can go ahead with Sentence Tokenizer and Paading method. In this section i will be using GSE for ease of implimentation and achieving target.
Distance will be calculated using the below menthod
Similarity:

Given two vectors A and B, the cosine similarity between them is calculated as:

$$\text{Similarity} = \frac{A \cdot B}{\|A\| \cdot \|B\|}$$

where $A \cdot B$ is the dot product of the vectors, and $\|A\|$, $\|B\|$ are the Euclidean norms of vectors A and B, respectively.

Distance:

$$\text{Distance} = 1 - \text{Similarity}$$

This represents that the higher the similarity, the smaller the distance between the vectors.

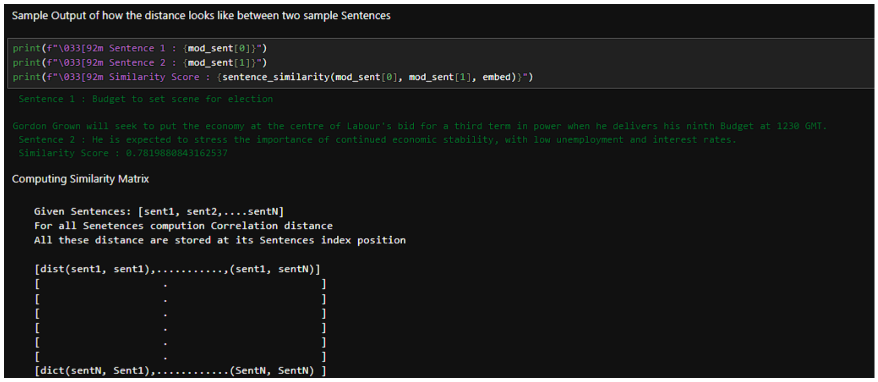


Figure 13: sample-Code

Summarization

The summary function combines multiple actions to extract the most relevant sentences from an article, effectively summarizing its content. The aim is to collect the top N most relevant sentences to represent the entire article.

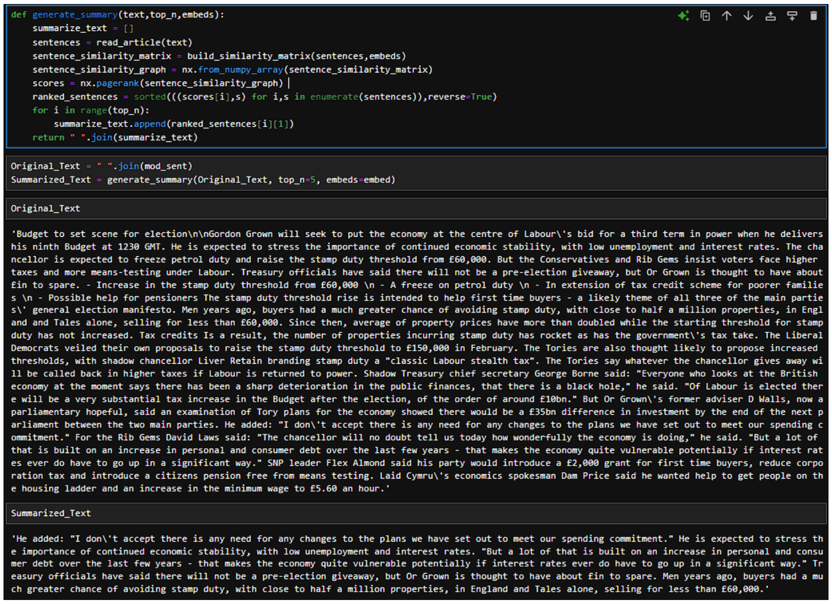
Steps:

1. **Reading the Article:** Extract text from the source document.
2. **Sentence Tokenization:** Split the article into individual sentence tokens.
3. **Compute Cosine Similarity:** Measure the similarity between each pair of sentence vectors using cosine similarity:

$$\text{Similarity}(A, B) = \frac{A \cdot B}{\|A\| \cdot \|B\|}$$

4. **Graph Representation:** Use the `NetworkX` library to build a graph where each node is a sentence and edges represent similarity scores.
5. **PageRank for Ranking Sentences:** Apply the PageRank algorithm on the graph to rank sentences based on their importance.
6. **Generate Summary:** Select the top N ranked sentences to construct the summary of the article.

Note: The steps outlined above are specific to the **Extractive Strategy** for text summarization.



```
def generate_summary(text,top_n,embeds):
    summarize_text = []
    sentences = read_article(text)
    sentence_similarity_matrix = build_similarity_matrix(sentences,embeds)
    sentence_similarity_graph = nx.from_numpy_array(sentence_similarity_matrix)
    scores = nx.pagerank(sentence_similarity_graph)
    ranked_sentences = sorted(((scores[i],s) for i,s in enumerate(sentences)),reverse=True)
    for i in range(top_n):
        summarize_text.append(ranked_sentences[i][1])
    return " ".join(summarize_text)
```

```
Original_Text = " ".join(mod_sent)
Summarized_Text = generate_summary(Original_Text, top_n=5, embeds=embed)
```

```
Original_Text
```

```
'Budget to set scene for election\n\nGordon Grown will seek to put the economy at the centre of Labour\'s bid for a third term in power when he delivers
his ninth Budget at 1230 GMT. He is expected to stress the importance of continued economic stability, with low unemployment and interest rates. The cha
ncellor is expected to freeze petrol duty and raise the stamp duty threshold from £60,000. But the Conservatives and Rib Gems insist voters face higher
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es ever do have to go up in a significant way." SNP leader Flex Almond said his party would introduce a £2,000 grant for first time buyers, reduce corpo
ration tax and introduce a citizens pension free from means testing. Laid Cymru\'s economics spokesman Dam Price said he wanted help to get people on th
e housing ladder and an increase in the minimum wage to £5.60 an hour.'
```

```
Summarized_Text
```

```
'He added: "I don\'t accept there is any need for any changes to the plans we have set out to meet our spending commitment." He is expected to stress th
e importance of continued economic stability, with low unemployment and interest rates. "But a lot of that is built on an increase in personal and consu
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ch greater chance of avoiding stamp duty, with close to half a million properties, in England and Tales alone, selling for less than £60,000.'
```

Figure 14: Code-Execution of summarization

6 Validation

There are multiple ways to compare two sentences to compute accuracy:

- **N-Grams / BLEU Score:** This method is mostly used in machine translation tasks. It compares the overlap of n-gram sequences between the predicted and reference sentences to measure the quality of generated text.
- **Similarity Score:** This method computes the semantic similarity between two sentences, often using cosine similarity on vector embeddings. It is primarily used in summary evaluation, similar word or sentence search, and semantic comparison.

In our case, the second option (Similarity Score) is more suitable. However, we will implement both methods to observe and compare the difference in their evaluation scores.

1.N-grams/Bleu Score:

$$BleuScore:0.3105109744864834$$

Conclusion:

We can Clearly see that score is only 31$perecent$ which does not means that summary is wrong since it is comparing the words but not the context and semantic meanings.

Hence even though both summary could mean the same and still Bleu Score will be less, and for the same very reason this comparision is used only for Translation purposes and not for this very perticular case.
2.Similarity Score:

Similarity score:0.5627625286579132

This gives us the better score comparitive to the Bleu Score for our use case i.e 56.3percent.

7 Conclusion

As we come to the end of the implementation, below is a summary of what has been done and achieved so far:

Summary

- Collected BBC articles along with their summaries to serve as reference and for comparison purposes.
- Explored multiple methods and techniques for text summarization, including both **Extractive** and **Abstractive** approaches.
- Performed a deep dive into various extractive methodologies.
- Chose the **Graph-based implementation method** for extractive text summarization.
- Converted the articles into sentence tokens.
- Computed a similarity matrix to build a graph where each node represents a sentence.
- Applied the **PageRank algorithm** to rank sentence tokens and selected the top N sentences to represent the final summary.
- For validation, implemented both **BLEU Score** and **Similarity Score** techniques. Concluded that BLEU Score is not suitable for this task, and the Similarity Score offers more reliable results in our context.

CODE LINK